STANDARD OPERATING PROCEDURE

Redd Master 1.0

Electronic data collection for spawning ground surveys

Field	Detail
SOP number	SGS-DC-01
Version	1.0
Effective date	_____
Applies to	Spawning ground survey crews and data staff
Supersedes	None — new procedure. Paper forms remain the fallback.
Application version	Redd Master 1.0 (schema reddmaster.sgs.v1)
Prepared by	Michael Stocklin
Reviewed by	_____ Date: _____
Approved by	_____ Date: _____
Review cycle	Annually, before the start of each survey season

READ BEFORE USE

Redd Master records survey data; it does not replace survey protocols. Follow the survey methods, safety requirements and sampling rules issued by the managing agency and co-managers. Where this SOP and an agency protocol disagree, the agency protocol governs.

1. Purpose and scope

This SOP describes how survey crews use the Redd Master application to record, check, and submit spawning ground survey data, and how that data reaches the program's data manager.

1.1 Scope

- Applies to all crews conducting spawning ground surveys using Redd Master on a phone or tablet.
- Covers the general survey (tally) form, the carcass form, the redd form, other-species observations, photographs, quality checks, export and submission.
- Does not cover survey methodology, carcass sampling technique, safety procedures, or escapement estimation. Those are covered by the program's field protocols and the annual pre-season meeting.

1.2 Relationship to the paper forms

Redd Master reproduces the fields of the 2026 Spawner Survey Data Form, the 2026 Chinook Carcass Form and the 2026 Chinook Redd Form. Crews may use the application in place of the paper forms where the field lead has approved it.

> **PAPER IS THE FALLBACK**
> Carry paper forms on every survey. If a device fails, is lost, or runs out of battery, revert to paper immediately and continue the survey. Do not abandon a survey because of a device problem.

2. Responsibilities

Role	Responsibility
Surveyor	Records data accurately in the field. Confirms the Recorded by name is their own. Resolves quality-check warnings before leaving the reach. Exports at the end of the day.
Field lead	Confirms devices are charged and the app is current before departure. Approves use of the app in place of paper. Decides when subsampling begins and notifies the group chat.
Data manager	Receives exports, verifies cross-checks, loads data to the program database, and confirms receipt to the crew.
Program lead	Owns this SOP, approves changes, and maintains the application and its deployment.

3. Definitions

Term	Meaning
Tally line	One row of the general survey form, covering one river mile.
Unsamp	A carcass from which no data can be collected — fish guts, spines, heavily scavenged fish. Counted only in Not sampled on the tally form.
HOR / NOR	Hatchery origin (ad clip and/or CWT present) / natural origin (neither).
PSM	Pre-spawn mortality — a carcass with spawn status 0%.
Subsampling	Recording mark status on all carcasses but fully sampling only 1 in N.
Draft	A record started but not yet saved. Kept automatically by the app.

Term	Meaning
Export	Writing survey data out of the app as a file for submission.
Recorded by	The person entering data, stamped onto each record for attribution.

4. Equipment and prerequisites

4.1 Required

- Phone or tablet with Redd Master installed and tested offline
- Device charged to 100% at departure; a battery pack for surveys over four hours
- Handheld GPS unit (Redd Master does not replace it — see 6.4)
- Paper forms, pencil and clipboard as fallback
- Standard sampling kit and safety equipment per the field protocol

4.2 One-time setup on each device

1. Install Redd Master following the program's installation instructions (INSTALL.md).
2. Open the app and confirm it loads.
3. Turn the device to airplane mode and open the app again. It must work normally. If it does not, do not take it into the field — report it to the program lead.
4. On the Survey page, under **Recorded by**, enter your initials and full name and press Save. This is stored on the device and only needs doing once.
5. Optional: Settings → Keep screen awake, to stop the device sleeping mid-survey.

> **ATTRIBUTION, NOT A PASSWORD**
> Recorded by identifies who entered the data. It is not a login and does not restrict access to the device or the data. Anyone holding the device can select any name. Treat it as a record of provenance for follow-up, not as proof of identity.

5. Before leaving the office

1. Confirm the device is charged and the app opens.
2. Confirm the app version in **Settings → Diagnostics** matches the version issued by the program lead.
3. Confirm any previous survey has been exported. The Survey page shows a warning if records exist that have not been exported.
4. Start a new survey with **Add survey** in the top bar. Crew, reach, stream and species are copied from the previous survey; confirm them. Do not add records to the previous day's survey.
5. Record the handheld GPS unit number you signed out.
6. Check flows and confirm the reach assignment and access with the field lead.

6. Field data collection

6.1 Survey header — complete on arrival

Complete every field marked with a red asterisk before recording observations. The app lists anything missing in the checks at the top of the Survey page.

Section	Entry
1 Survey identification	Target species; fork (NF / MF / SF / Other); stream name if Other; Recorded by; crew initials; date; reach; water type (Mainstem or Tributary); for a tributary, which one

Section	Entry
	(Peat Bog Creek, Clearwater Creek or Other); survey type (Primary or Braid)
2 Survey conditions	Visibility 1–6, judged on both turbidity and glare
3 Upper and lower readings	Time, water temperature in °C (a °F box beside it converts automatically) and river mile at the upper and lower ends of the reach. Use Set now for the time.
4 Subsampling and page	Subsampling Y/N and rate if used; handheld GPS unit number; notes-on-reverse; page number

WATER TYPE DRIVES THE CALCULATION

Mainstem NF and MF use carcasses × expansion and a 7–10 day survey cycle. Tributaries such as Peat Bog use carcasses recovered with no expansion and are surveyed as often as possible. SF uses redds × 2.5. Setting Water type incorrectly will produce the wrong method on the Escapement card.

6.2 Report — page 1 builds itself

1. Open the Report page. River miles appear on their own as redds and carcasses are saved with a river mile; use Add river mile, to the nearest tenth, for a mile with only live fish or nothing seen.
2. On each river mile line, tap + for live fish not on a redd and for unsampled dead. New redds, sampled carcasses and live fish on redds are counted automatically from the Redds and Carcass pages (see 6.3 and 6.4), so count fish on a redd on the redd form only.
3. Unsampled dead are the only dead fish entered by hand. The columns below are what the Report builds for page 1.

Column	Use for
No subsampling — Sampled	Carcasses worked up on the carcass form
No subsampling — Not sampled	Unsamps only: guts, spines, heavily scavenged fish. No carcass record is made.
Subsample group — Sampled	Fully sampled fish pulled from a subsample lineup
Subsample group — No biologic samples	Subsampled fish where mark status was recorded but no biological samples taken

4. Add notes for other species, anything uncertain, or unusual observations.
5. Add a line for every river mile walked, even where nothing was seen, so page 1 shows the full reach.

6.3 Carcasses

1. Press **Add carcass** on the Carcass page.
2. Complete sections 1–4: species, otolith and DNA numbers, fork length in cm, sex, river mile, percent spawned, ad clip, CWT, sample numbers, scavenged, eyes missing.
3. Record mark status on every carcass, including those not fully sampled.
4. Capture photographs (section 5) where the fish is unusual, disputed, or tagged.
5. Use **Save + next** when working a pile of carcasses. Sample numbers advance automatically and species is pre-filled from the survey. Each saved carcass of the target species is counted on the Report page at its river mile. A carcass saved without a river mile is flagged there until one is added.
6. Cut the tail when the fish is finished, per the sampling protocol.

MEASURE ONLY INTACT FISH

Do not record fork length unless the body is intact from head to tail. Never estimate a length.

6.4 Redds

1. Press **Add redd**. Redd ID starts at 01 each day and increases. Each saved redd, and its live fish, is counted on the Report page at its river mile.
2. Take a new waypoint on the handheld GPS for every redd, even where redds are close together, and enter the waypoint number.
3. Record channel type, braid location if braided, and location in channel.
4. Record the number of live Chinook on the redd.
5. Flag the redd with date, number of live fish, initials and distance from the flag.

TWO GPS RECORDS, AND THEY ARE NOT INTERCHANGEABLE

The handheld waypoint number plus the GPS unit number is the record of truth for redd position. The coordinates the app captures from the device are supplementary. Record both.

6.5 Other species

Record live fish, carcasses or redds of any species other than the survey target on the Other page. This keeps the primary counts clean while preserving the observation.

6.6 Photographs

- Photographs may be attached to any carcass, redd or other-species record.
- Images are reduced automatically for storage. Take them close and square to the subject.
- Include a scale reference where size matters.
- Photographs are stored on the device and exported separately (see 8.3).

7. Quality checks in the field

Redd Master runs two reconciliation checks continuously. Both must pass before the crew leaves the reach. They appear on the Report page and again at the end of the survey.

Check	Rule	If it fails
Redds	New redds tallied = number of redd form records	A redd was tallied but not recorded, or recorded twice. Find and correct before leaving.
Carcasses	Sampled dead columns (three) = number of carcass records for the target species	A carcass was tallied but not worked up, or an unsamp was put in the wrong column.

7.1 Other warnings

- Required field empty — the record cannot be considered complete.
- Value outside the expected range, for example a fork length below 25 cm or above 160 cm.
- Duplicate otolith, DNA, HBL, line, redd ID or waypoint number, naming the record that already uses it.
- CWT recorded as detected but no head bag label number entered.
- No recorder set on the survey.

> **WARNINGS DO NOT BLOCK SAVING**
> The app will let you save a record with warnings, because field conditions sometimes require it. Anything left unresolved must be explained in the notes so the data manager can interpret it.

8. End of survey

8.1 Close out

1. Enter the lower time, temperature and river mile in the survey header.
2. Open the survey menu and select **Finish the survey**.
3. Resolve any failed checks. Confirm no unsaved drafts remain.
4. Review the day totals against your recollection of the survey.

8.2 Export the data

1. From Finish the survey, or from Export & backup, download the **Excel workbook**.
2. Where the program loads data to a database, also export **Relational JSON** or the **SQL insert script** as directed by the data manager.
3. Send the export to the data manager by the method the program has agreed.
4. Copy the day summary and post it to the group chat and the office whiteboard.

8.3 Photographs

1. Open **Photo archive** and select **Export photos**. This produces a ZIP with an index that links each image to its record.

8.4 Back up

1. Run **Back up everything** at least weekly and always before a device is wiped, updated or handed over. Store the backup file where the program keeps its field records.

DATA LIVES ONLY ON THE DEVICE UNTIL IT IS EXPORTED

There is no automatic upload. An unexported survey exists on one phone and nowhere else. Export at the end of every survey day, without exception.

9. Data submission and records

Item	Format	When	To
Survey data	Excel workbook	End of each survey day	Data manager
Database load	Relational JSON or SQL	End of each survey day, if required	Data manager
Photographs	ZIP with index	End of each survey day where photos were taken	Data manager
Day summary	Text	On return to the office	Group chat and whiteboard
Full backup	JSON	Weekly and before any device change	Program records

The data manager confirms receipt. Do not clear data from a device until receipt has been confirmed.

10. Troubleshooting

Problem	Action
A form was closed by accident	Nothing is lost. The draft is kept — an amber Unsaved draft bar appears on that page. Press Resume.
The app will not load or looks wrong	Settings → Diagnostics → Force update, then reopen. If it persists, use paper and report it.
An error banner appears	Settings → Diagnostics → Copy report and send it to the program lead. Continue on paper if the app is unusable.
Photos will not attach	The device is blocking the photo store, usually private browsing or a local file copy. Continue without photos and report it.
GPS will not fix	Move to open sky. If it still fails, enter coordinates by hand or rely on the handheld waypoint and river mile.
Device lost, broken or flat	Revert to paper immediately. Report to the field lead the same day. Any unexported data on that device is lost.
Wrong data saved	Tap the record to edit it. Deleted records and drafts can be restored with Undo for a few seconds after deletion.

11. Limitations and cautions

- The escapement figure shown in the app is a field running total for situational awareness. It is not the official estimate. The office applies the final expansion and quality assurance.
- Recorded by is attribution, not authentication. It does not verify identity.
- Data is stored on the device only. Clearing the browser's site data, uninstalling, or losing the device will destroy unexported records.

- The app records what is entered. It does not verify field method, species identification or sampling technique.
- Photographs consume device storage. Export and review storage use weekly.

12. Revision history

Version	Date	Summary of change	Approved by
1.0	_____	Initial issue for Redd Master 1.0	_____

Appendix A — Field codes

Visibility

Code	Meaning
1	Excellent
2	Very good
3	Good
4	Fair
5	Poor
6	Not surveyable

Carcass fields

Field	Values
Sex	M · F · UNK
Percent spawned	0 · 50 · 100
Ad clip	Y · N · Unk
CWT	Y · N · Unk
Eyes missing	0 · 1 · 2
Origin (derived)	HOR = ad clip and/or CWT · NOR = neither · UNK = either unknown

Redd fields

Field	Values
Location in channel	LB left bank · MC mid channel · RB right bank
Channel type	Single · Braided
Braid location	L · M · R (more than three braids: describe in notes)

Appendix B — Mark status decision

Observation	Record
Wand beeps	CWT = Y
Tag subsequently lost	CWT = Y, and note it
No beep	CWT = N
One or two eyes missing and no beep	Not wandable — CWT = Unk

Collect a snout only where a CWT is detected. Wand again after the snout is placed in the head bag, and record the head bag label number.

Appendix C — Escapement methods

Reach type	Method	Survey frequency
North Fork and Middle Fork, mainstem	Carcasses recovered × expansion	Every 7–10 days
Middle Fork tributaries (e.g. Peat Bog)	Carcasses recovered = escapement	As often as possible, to clear carcasses
South Fork	Redds observed × 2.5	Per survey schedule

For the South Fork, carcasses determine the proportion of each stock or origin. That proportion applied to total escapement gives the escapement for each origin.

> **FIELD FIGURES ARE PROVISIONAL**
> The expansion factor is issued by the office and is not derived in the field. Figures shown in the app are for situational awareness only.

Appendix D — Pre-spawn mortality subsampling

Begin subsampling when it is unlikely all carcasses in a reach can be sampled — typically 50–100 carcasses. Start at a rate of 1 in 10.

- Notify the group chat when your crew starts subsampling, and again if the rate changes to 1 in 20.
- Subsample hatchery origin fish with spawn status 0% only.
- Fully sample natural origin fish, unknown origin fish, and hatchery fish with spawn status above 0%.
- Record mark and tag status for every fish in the lineup.
- Do not select which fish to sample. Take them in order — choosing introduces bias.
- If a pulled fish cannot be sampled, take the next fish in order.
- Subsampling restarts by reach or by day. Not reaching the tenth fish is expected and acceptable.

The Reference page in the app contains this workflow, a decision helper and a lineup counter.